

Project 2 LLM Transcript

Course: DATA 607 – Project 2

10/10/2026

LLM Transcript and Citation

Tool Used: ChatGPT (OpenAI, GPT-5.2)

Purpose: Assistance with designing reproducible data tidying pipelines, validating wide-to-tidy transformations using `pivot_longer()`, structuring column normalization logic, and refining analytical explanations for visualizations.

Overview of LLM Usage

I used ChatGPT primarily as a **conceptual assistant for data transformation design and analysis interpretation**. The model helped guide several steps of the project:

1. Designing wide-to-tidy transformation pipelines using `pivot_longer()`.
 2. Structuring column normalization when multiple variables were embedded in column names.
 3. Improving the clarity of explanations for plots and analytical interpretations.
- All code suggestions were tested locally in RStudio and adjusted when necessary before inclusion in the final Quarto document.

1

[1] Conversation with ChatGPT (Wide-to-Tidy Transformation Strategy)

Prompt:

How should I convert a dataset where multiple measurements are stored as separate columns into tidy format using `tidyr`?

Assistant (ChatGPT):

Use `pivot_longer()` to reshape the dataset so that measurement names become values in a new variable column and the corresponding observations become values in another column.

Example structure:

```
tidy_data <- raw_data |>
  pivot_longer(
    cols = -id,
    names_to = "variable",
    values_to = "value"
  )
```

This approach ensures that:

- Each variable becomes a column.
- Each observation becomes a row.
- Each value occupies one cell.

[2] Conversation with ChatGPT (Column Normalization from Encoded Variables)

Prompt:

How should I separate column names like `DE_wind_capacity` into separate variables representing country and energy type?

Assistant (ChatGPT):

When multiple variables are encoded inside column names, you can first reshape the dataset

with `pivot_longer()` and then extract the embedded components using regular expressions.
Example workflow:

2

```
pivot_longer(cols = -day, names_to = "country_energy", values_to = "capacity") |>
extract(
  country_energy,
  into = c("country_code", "energy_type"),
  regex = "^(.*)_(.*)_capacity$"
)
```

This converts information previously embedded in column names into separate variables, allowing the dataset to follow tidy data structure.